

Energy Harvesting over Optical Communication Networks Using Discarded Light from a Gain-Flattening Filter

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Abstract We demonstrate electrical-domain energy harvesting over optical communication networks and show for the first time that more than 10 mW of electrical power per core can be extracted from discarded optical power at a gain-flattening filter in a four-core erbium-doped fibre amplifier. ©2026 The Author(s)

Introduction

Energy harvesting (EH) enables the effective reuse of small amounts of energy available in the surrounding environment [1, 2]. For example, in a fibre Bragg grating (FBG)-based temperature sensing system, 6 mW of electrical power has been recovered from residual amplified spontaneous emission (ASE) used as a light source [2].

The optical power handled in optical communication networks (OCNs) continues to increase with growing traffic demands. However, not all injected optical power reaches the receiver as useful signal power; part of it is lost through transmission loss, reflections, or spectral components intentionally removed by system functions. Such discarded light has conventionally been treated as an unavoidable by-product of system operation. Therefore, EH over OCNs is expected not only to improve the utilization efficiency of injected optical power but also to provide a potential power source for low-power devices and sensors deployed in or around the network.

Prior research [3, 4] has reported the reuse of optical pump power in erbium-doped fibre amplifiers (EDFAs), which can be regarded as EH over OCNs in the optical power domain. The conversion of residual ASE into electrical power has also been demonstrated in distributed fibre-optic sensors and optical sensing systems [2]. In contrast, electrical-domain EH that directly converts light discarded by system functions inside communication network equipment into electrical power for device operation has not yet been achieved.

In this study, we investigate electrical domain EH over OCNs by focusing on light rejected by a gain-flattening filter (GFF). We propose a highly efficient recovery method and, to the best of our knowledge, succeeded for the first time in obtaining more than 10 mW of electrical power per core. The proposed method is beneficial for accelerating the convergence of data communication and network sensing.

Four-core EDFA with an aggregated power extraction GFF

In EDFA-based transmission systems, a GFF is essential for suppressing wavelength-dependent gain deviations. In general, optical power removed by the GFF is dumped so that it does not affect communication quality. We propose an EH scheme that recovers discarded optical energy and converts it into electrical power by means of a photovoltaic power converter (PPC) for device operation. Since multicore EDFAs [5–8] are promising candidates for amplification in OCNs, we consider a four-core (4C) EDFA as a case study. Figure 1 shows the configuration of a four-core EDFA (4C-EDFA) incorporating the proposed aggregated power extraction GFF (APE-GFF). The APE-GFF is fully compatible with conventional GFFs and can be installed at the amplifier output stage.

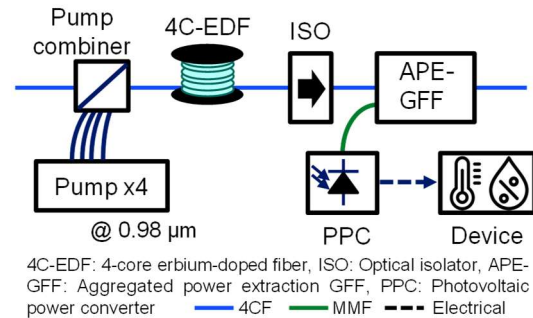


Fig. 1: 4C-EDFA incorporating the APE-GFF.

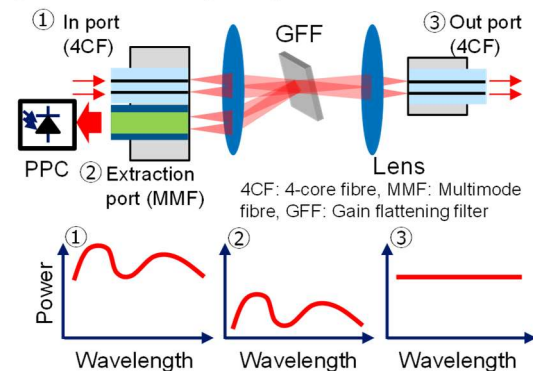


Fig. 2: Configuration and operating concept of the APE-GFF in a 4C-EDFA.

Figure 2 illustrates the configuration and operating principle of the APE-GFF, where gain equalization in the four-core EDF is achieved while light rejected by the GFF is guided to an extraction port and converted into electrical energy by a downstream PPC. Although the input port supports multiple spatial paths via a four-core fibre (4CF), aggregating the extracted light into a single MMF simplifies the PPC configuration after power recovery. The optical spectra at the respective ports are denoted as 1–3 in Fig. 1. In this study, we implemented the GFF using a thin-film filter (TFF), as the rejected light is spatially separated by reflection and more easily extracted than in grating-based approaches such as long-period gratings.

We designed the MMF for the extraction port such that the input 4CF had a square-lattice core arrangement with a 40- μm pitch and an 8.5- μm mode-field diameter (MFD) at a wavelength of 1310 nm. Figure 3 shows the coupling efficiency between the 4CF and the MMF as a function of MMF core diameter for an equal-magnification optical system. Even with this simple optical system, full optical power can be received for core diameters of 70 μm or larger. Therefore, we selected a commercially available MMF with a 105- μm core diameter and an NA of 0.2.

Finally, the electrical power obtainable from a single 4C-EDFA was numerically estimated as a function of gain tilt, which determines the amount of light removed by the GFF. The PPC optical-to-electrical conversion efficiency was assumed to be 40%, based on the characteristics of the device (AFBR-POC205A9; Broadcom inc.) used in the experiment. Eight signal wavelengths from 1530 to 1565 nm with 5-nm spacings were assumed, with an input power of -9 dBm per wavelength and an output power after the GFF of 8 dBm per wavelength (17-dB gain). We evaluated the dependence of the recoverable power on gain tilt as defined by Eq. (1), where $\text{Gain}(\lambda_1)$ and $\text{Gain}(\lambda_2)$ denote the gains at 1530 nm and 1565 nm, respectively, and the bandwidth is 35 nm.

$$\text{Gain tilt (dB/nm)} = \frac{\text{Gain}(\lambda_1) - \text{Gain}(\lambda_2)}{\text{Band width}} \quad (1)$$

Figure 4 shows gain spectra for positive and negative gain tilts. In each case, the EDF length was optimized according to the gain tilt, and the pump power was adjusted such that the minimum gain was maintained at 17 dB. Figure 5 shows the gain tilt dependence of the pump-to-signal power conversion efficiency (PCE) and the recoverable electrical power. In the region where $|\text{Gain tilt}| > 0.05$, the PCE deteriorates because excessive pump power is required relative to the signal output, which is unsuitable for amplifier operation.

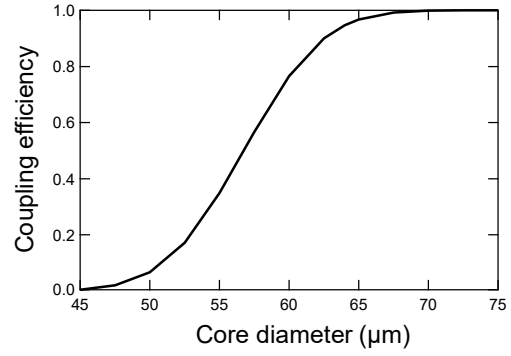


Fig. 3: Coupling efficiency between the 4CF and the MMF as a function of the MMF core diameter.

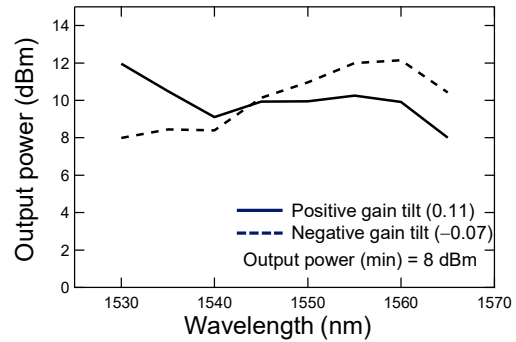


Fig. 4: Gain spectra of the four-core EDFA for two gain-tilt conditions (positive and negative gain tilt).

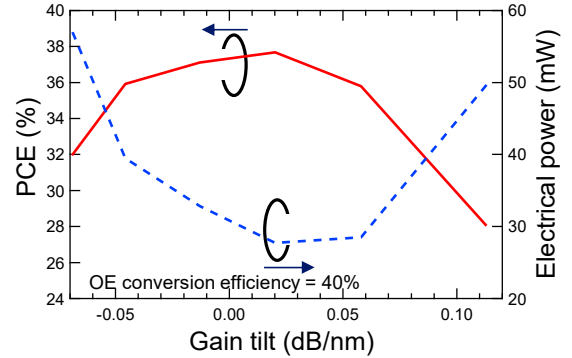


Fig. 5: Dependence of PCE and recoverable electrical power on gain tilt.

In contrast, even in the practical region where $|\text{gain tilt}|$ is below 0.05 and no remarkable PCE degradation is observed, a total electrical power of approximately 30–40 mW remains recoverable across all four cores. This indicates that a non-negligible amount of power would otherwise remain unused even under practical amplifier operating conditions.

Experimental results

Next, we evaluated the amplification characteristics and recoverable electrical power using a 4C-EDFA incorporating the proposed APE-GFF. As shown in Fig. 2, the setup included a 4C-EDFA comprising a pump combiner, four 0.98- μm pump lasers, a 4C-EDF [8], an optical isolator, the APE-GFF, and a PPC. The fabricated APE-GFF

exhibited a 93% extraction efficiency from input to extraction port. The signal light consisted of eight wavelength channels from 1530 to 1565 nm at 5-nm spacings and -9 dBm per wavelength. Amplification characteristics were measured with an optical spectrum analyzer, and electrical power with an electronic load. Figure 6 shows the average gain and NF spectra over all cores with and without the APE-GFF. Without the GFF, the average gain was 18.5 dB with a maximum variation of 2.9 dB. Introducing the GFF reduced the gain ripple to 1.4 dB, confirming effective gain flattening. The remaining gain variation is attributed to excessive attenuation above 1545 nm and is expected to improve with further optimization of the GFF characteristics. The GFF insertion loss, including MCF connection loss, was about 0.7 dB, which is comparable to commercially available single-mode GFFs. No NF degradation was observed, indicating a limited impact of the APE-GFF on EDFA performance. The black and red solid lines in Fig. 7 respectively denote the current and corresponding power characteristics versus voltage under the amplification conditions in Fig. 6. At the optimum load, 70 mW of electrical output was obtained, while the extracted optical power was 162 mW. The optical extraction, PPC conversion, and overall efficiencies were 93%, 43%, and 40%, respectively. Because the present APE-GFF introduced excessive attenuation, the obtained power was higher than expected. The red dashed line indicates that about 45 mW can be obtained from a single 4C-EDFA under proper gain equalization, which is in reasonably good agreement with the calculated result in Fig. 5. More than 10 mW of electrical power was obtained per core, which to our knowledge is the highest power reported so far among EH studies.

Finally, using the setup shown in Fig. 8(a), we conducted a temperature and humidity sensor driving experiment in which optical power from the four-core EDFA served as the energy source. The PPC output supplied the sensor control circuit and humidity sensor. Figure 8(b) shows the measured temperature and humidity. The system was sampled every 10 s, and continuous operation for 10 min was confirmed.

The results showed that a single 4C-EDFA can provide 45 mW of electrical power, corresponding to an aggregated power of approximately 1.4 W in a submarine cable with 16 fibre pairs. This power is promising for driving low-power subsea sensing nodes. In SMART cables [9] equipped with subsea observation sensors, target observables include temperature, ocean-bottom pressure, and seismic motion, and high reliability and functional independence from the telecommunications system are required.

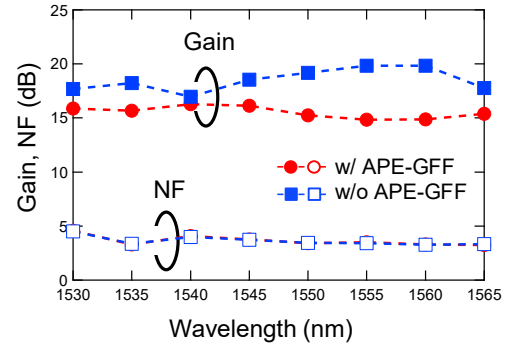


Fig. 6: Gain and noise figure (NF) spectra with and without the APE-GFF.

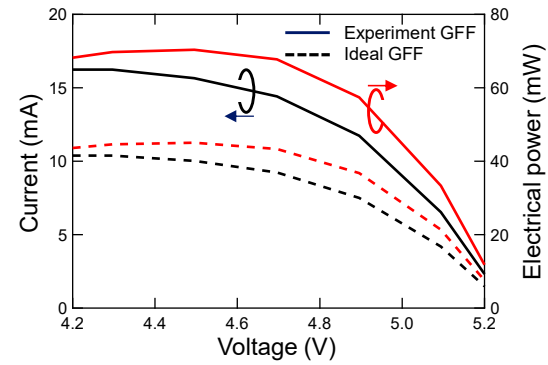


Fig. 7: Current-voltage characteristics of the recovered electrical power.

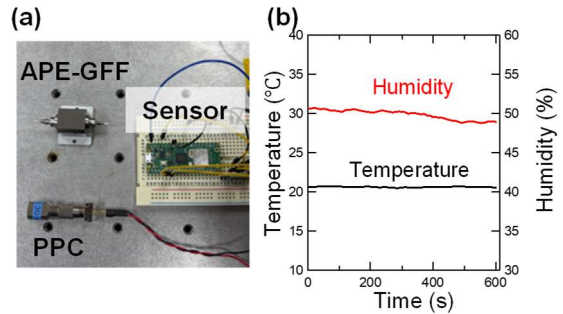


Fig. 8: (a) Measurement setup and (b) results of temperature and humidity.

Since a broadband ocean-bottom seismometer combined with a low-power digitizer typically operates at the few-hundred-milliwatt level [10], an obtained power of approximately 1.4 W is expected to be useful for driving multiple sensing nodes.

Conclusions

We demonstrated EH over OCNs using a 4C-EDFA with an aggregated power extraction GFF. The proposed APE-GFF achieved 93% extraction efficiency of discarded light and 40% overall energy recovery efficiency without degrading the gain or noise figure. Under suitable gain equalization, more than 10 mW per core and about 45 mW from a single 4C-EDFA were obtained, showing strong potential to power low-energy sensing nodes in future space-division-multiplexed fibre networks.

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